

UNSUPERVISED CLUSTERING TO IDENTIFY ZINC SOAP DEGRADATION IN OIL PAINT

Elise M. Etrheim, Materials Science & Engineering



***Scenes from the Life of
Saint John the Baptist***

Francesco Granacci

(Italian, ca. 1506-7)

Tempera, oil, and gold on
wood

Metropolitan Museum of Art,
1970.134.1



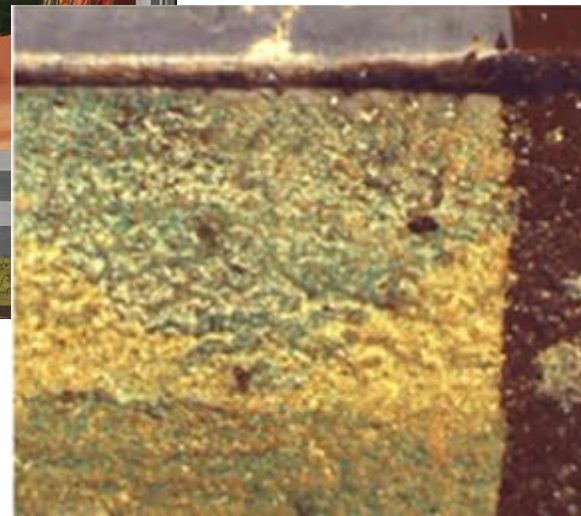
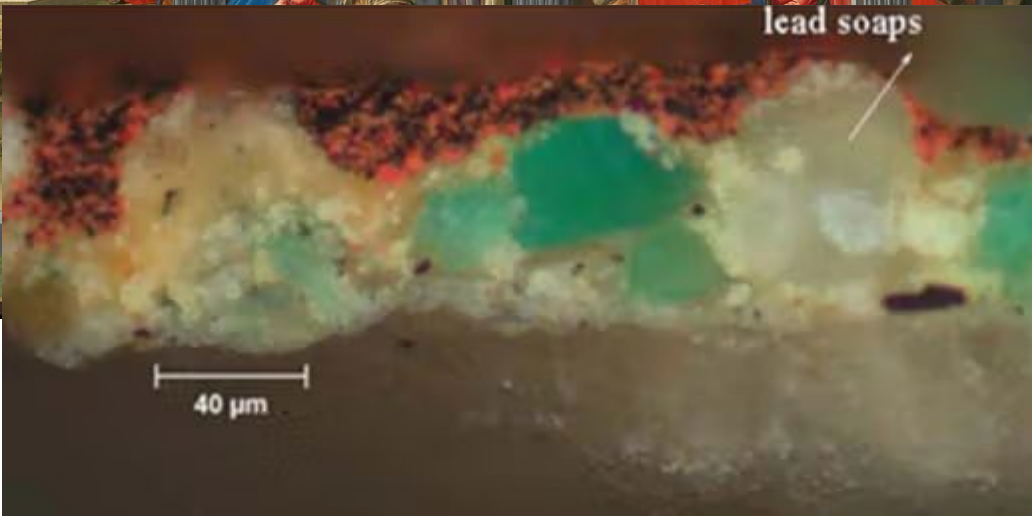
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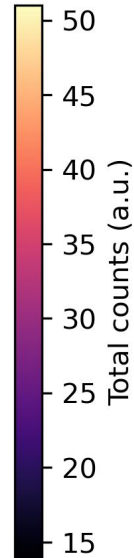
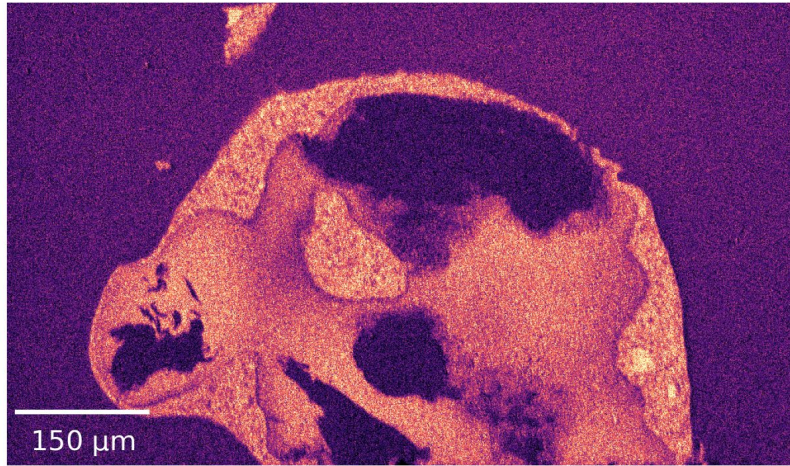
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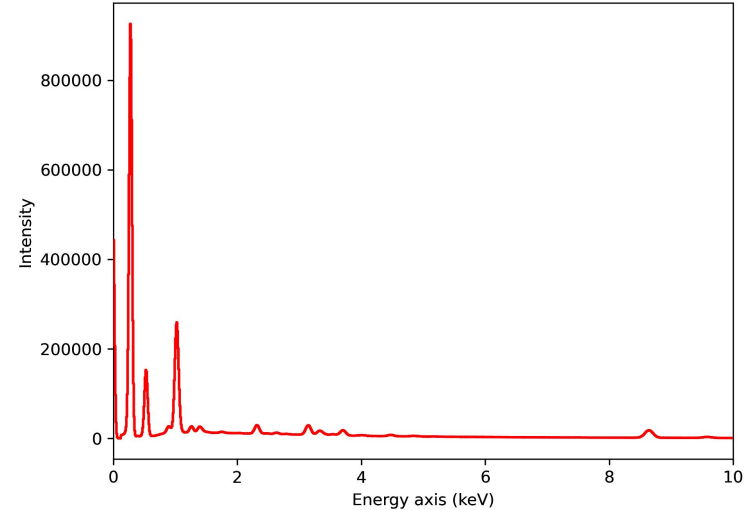
Motivation

Hypothesis: Unsupervised clustering of SEM-EDS elemental data will identify spatially coherent, chemically distinct regions associated with zinc soap formation in oil paint samples

Aerial: total counts map



Aerial: summed EDS spectrum (view 0-10 keV)



The Dataset

- SEM-EDS datasets
- The dataset comes from a publicly available NIST repository accompanying a zinc soap paint study
- Examining a zinc soap protrusion observed on an artificially and naturally aged oil paint test panel

STUDIES IN CONSERVATION
2020, VOL. 65, NO. 1, 14–27
<https://doi.org/10.1080/00393630.2019.1666467>



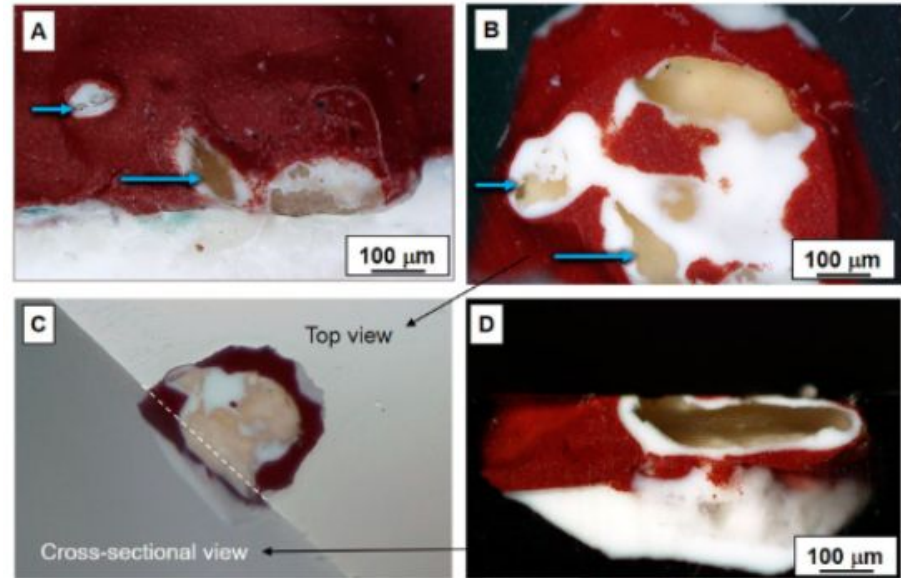
Routledge
Taylor & Francis Group

ORIGINAL RESEARCH OR TREATMENT PAPER

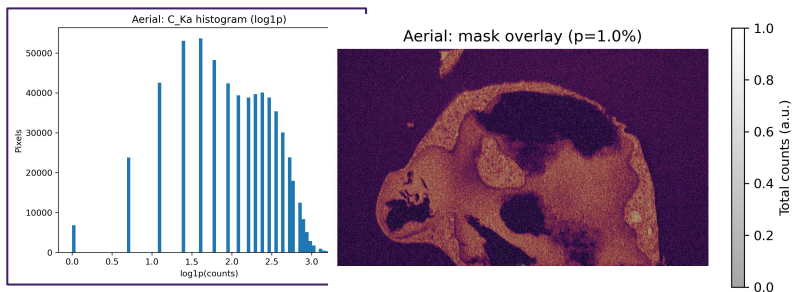
Characterization of Zinc Carboxylates in an Oil Paint Test Panel

Christine Romano¹, Thomas Lam¹, G. Asher Newsome ¹, Joshua A. Taillon ², Nicole Little ¹ and Jia-sun Tsang¹

¹Museum Conservation Institute, Smithsonian Institution, Suitland, MD, USA; ²Material Measurement Laboratory, National Institute of Standards and Technology, Gaithersburg, MD, USA



Workflow



Elemental
Feature
Extraction

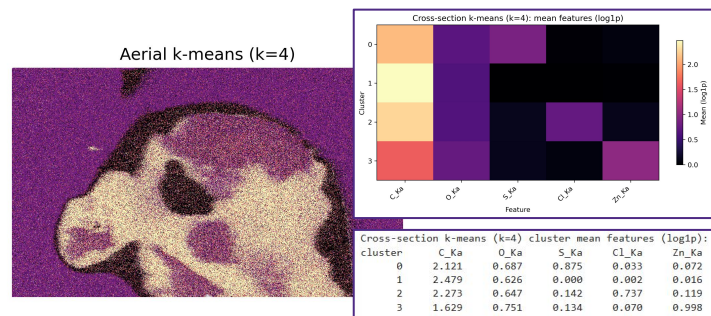
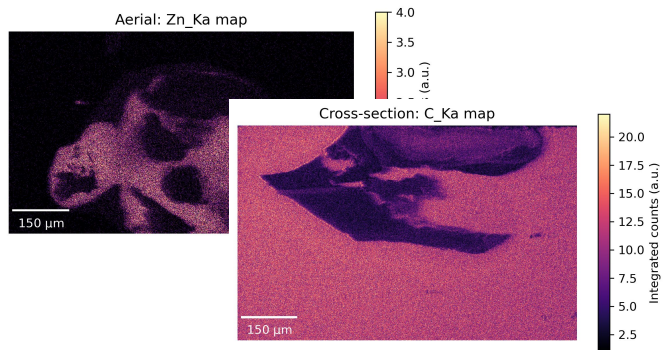
Masking of
Low-Signal
Pixels

Feature
Log
Transform

K-Means

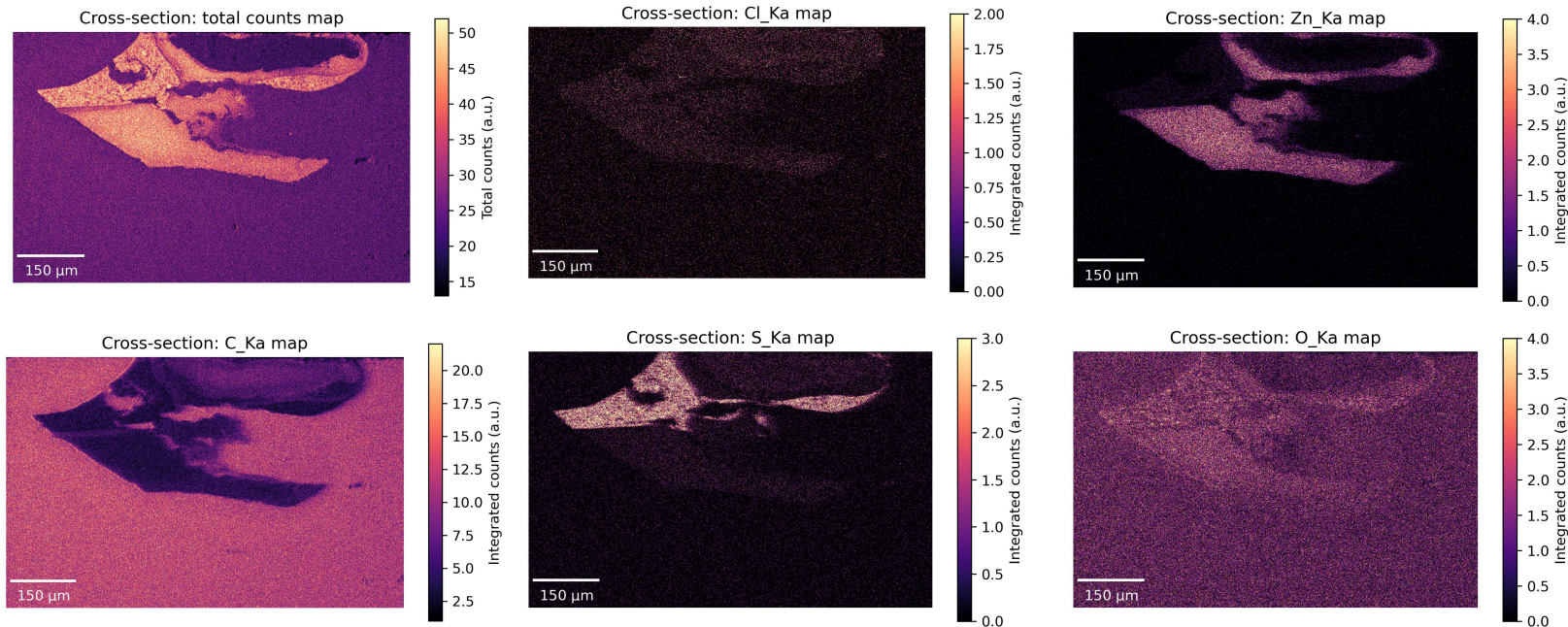
GMM

Hierarchical



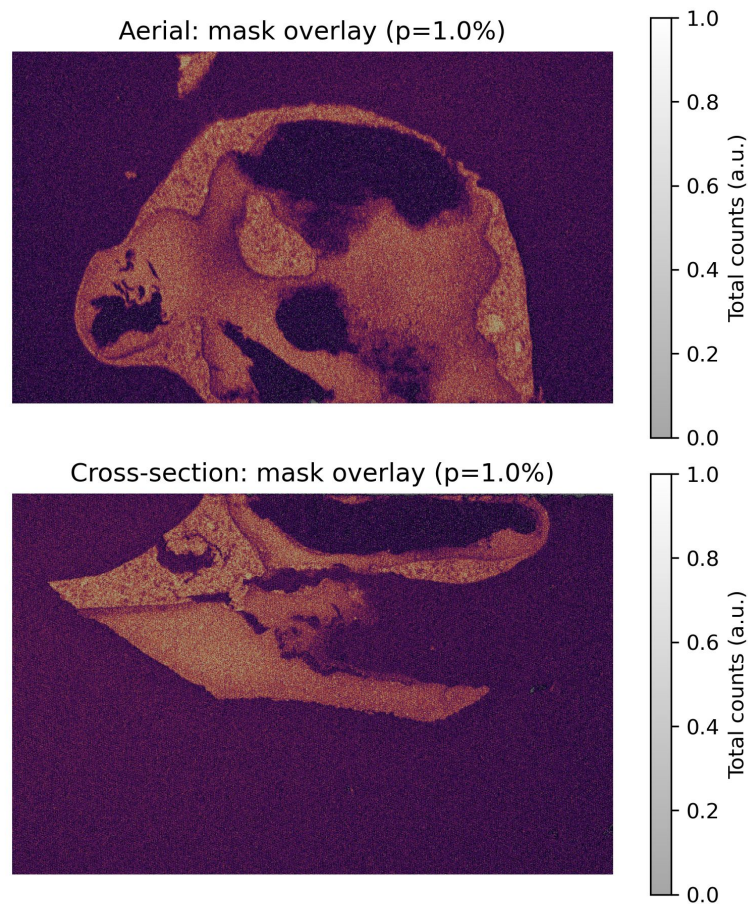
Elemental Feature Extraction

- Characteristic X-ray emission peaks integrated to obtain elemental intensity features for each pixel



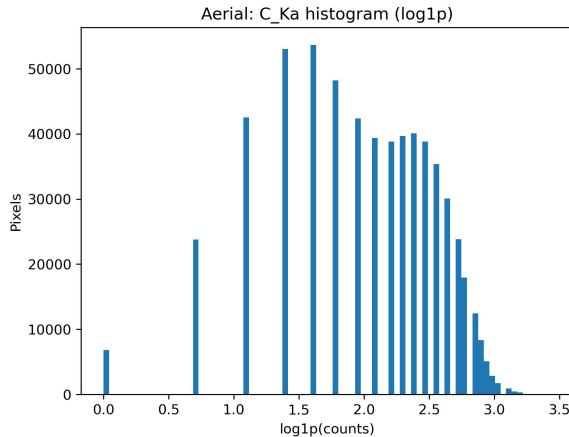
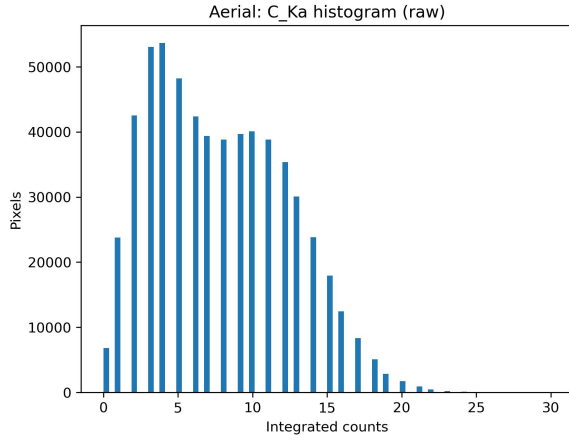
Data Cleaning & Masking

- Pixels with extremely low total X-ray counts were removed using a 1% intensity threshold
- These pixels typically correspond to vacuum, detector noise, or acquisition artifacts
- Prevents clustering from being driven by noise rather than composition



Feature Transformation & Standardization

- Elemental intensities transformed using $\log(1+x)$
- Compresses dynamic range and reduces dominance of extreme values
- Standardized so that high-intensity elements would not dominate clustering



Clustering Methods

1. K-means

- assigns points to nearest cluster centroid and iteratively updates centroids to minimize within-cluster variance

2. GMM

- models the data as a mixture of Gaussian distributions and estimates probability that each point belongs to each cluster

1. Hierarchical

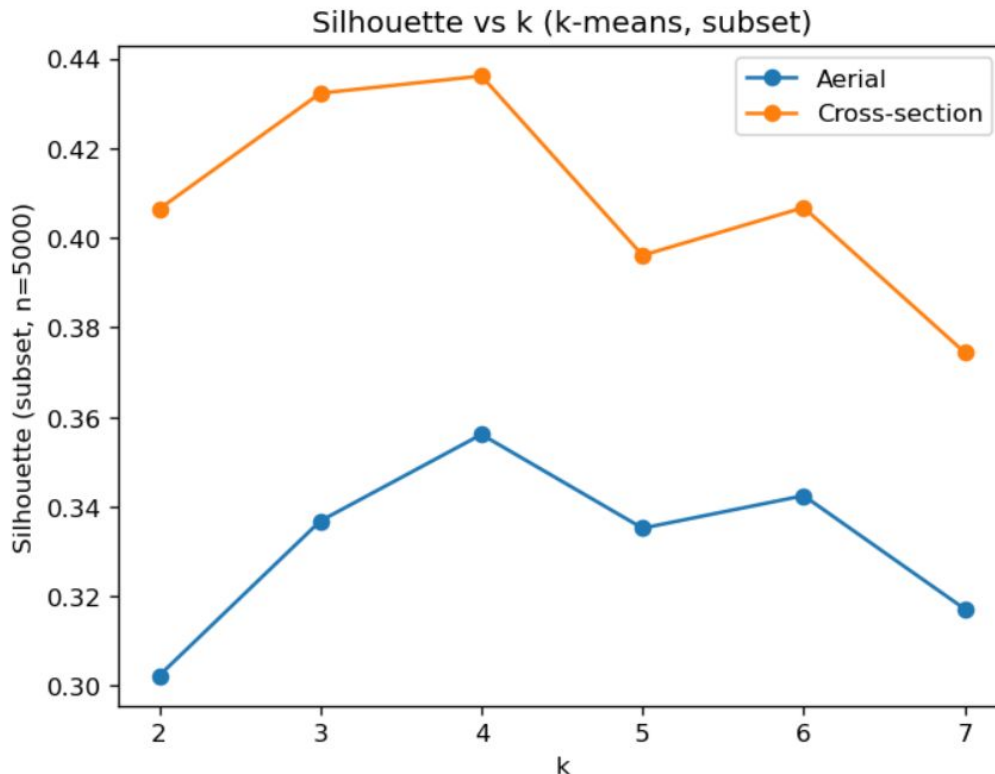
- builds a tree of clusters by iteratively merging or splitting groups based on similarity

Determining a number of clusters (k)

- Silhouette score

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

- Evaluated Calinski-Harabasz score and BIC
 - Sil: k = 4
 - CH: k = 5
 - BIC: k > 7
- Qualitative cluster comparison guided selection of k = 4

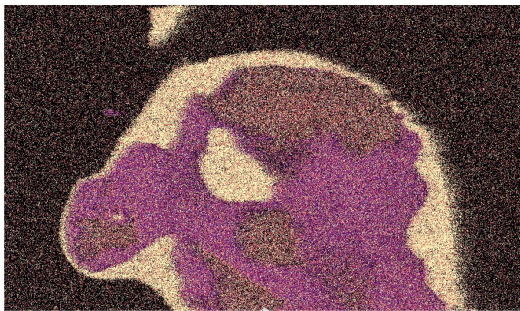


Cluster Maps

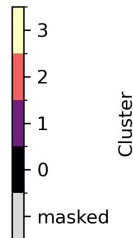
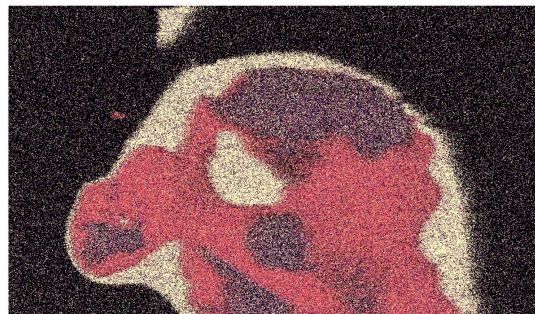
Aerial k-means (k=4)



Aerial GMM (k=4)



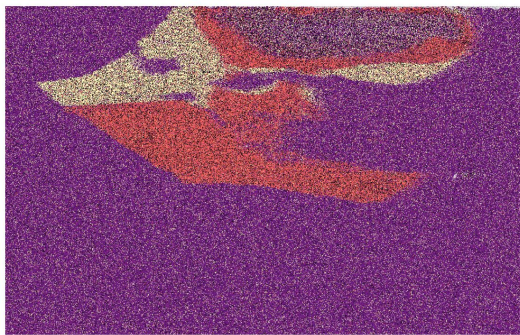
Aerial hierarchical (subset-fit, k=4)



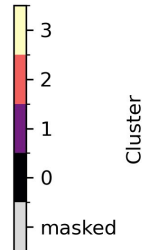
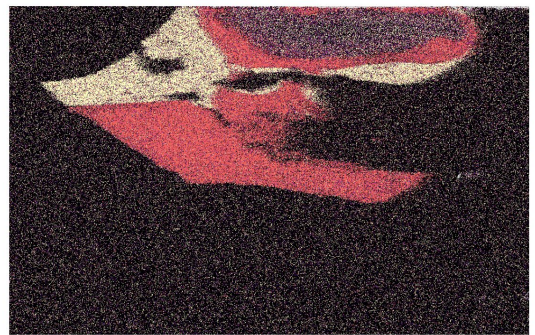
Cross-section k-means (k=4)



Cross-section GMM (k=4)

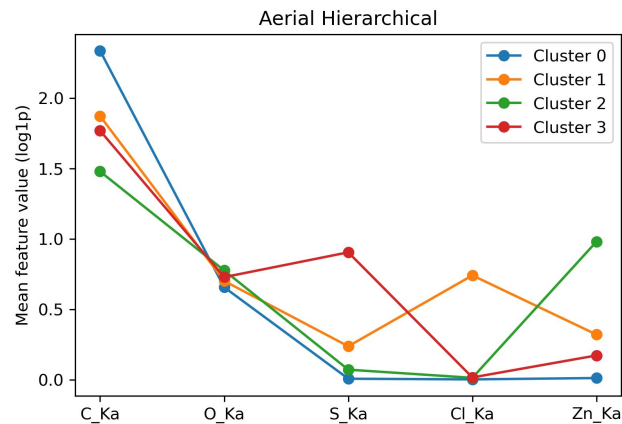
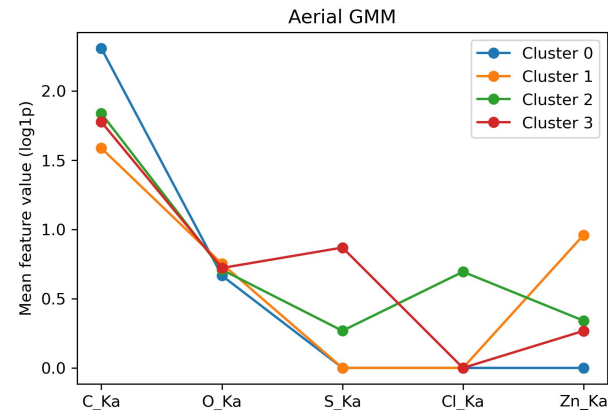
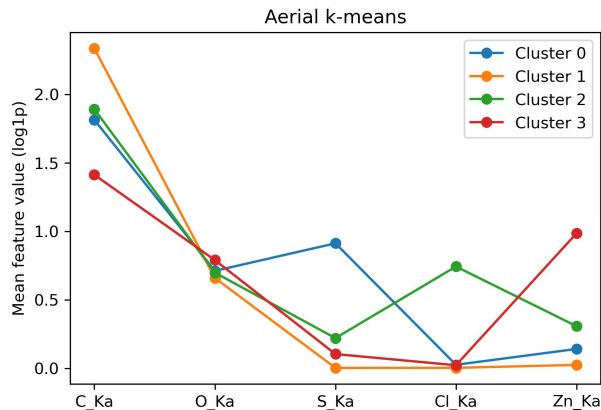


Cross-section hierarchical (subset-fit, k=4)



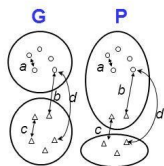
Cluster chemistry

- Mean elemental features within each cluster
- Four consistent signatures:
 - Zn-rich regions
 - Cl-rich regions
 - S-rich regions
 - and a matrix-dominated region



Method comparison

- Adjusted Rand Index (ARI)

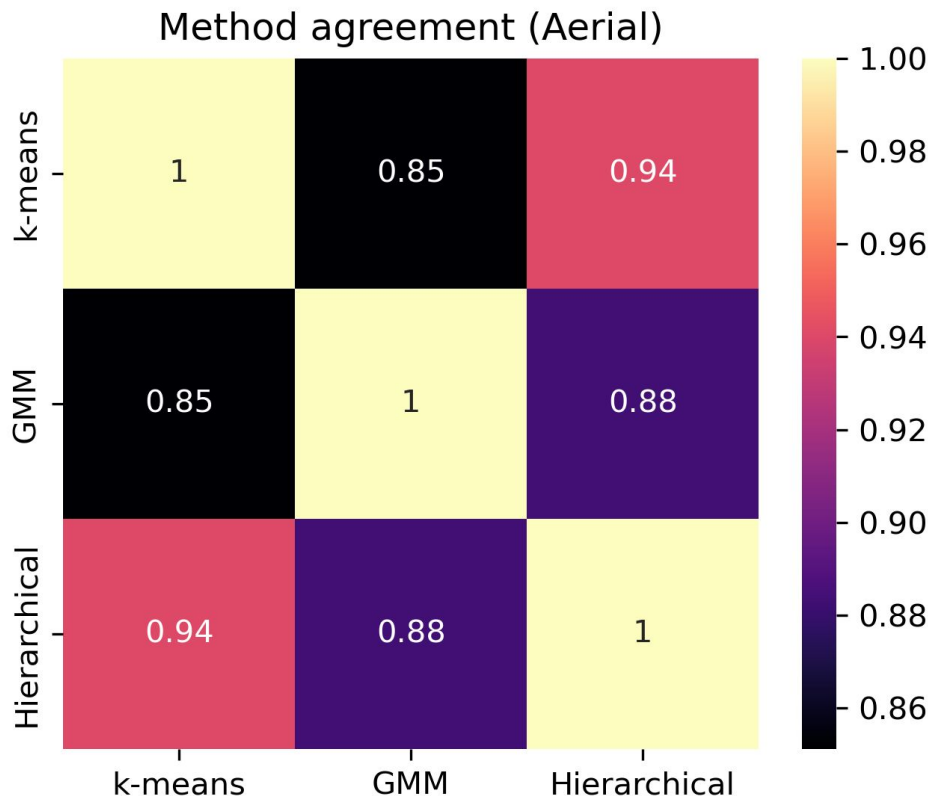


Agreement: a, d
Disagreement: b, c

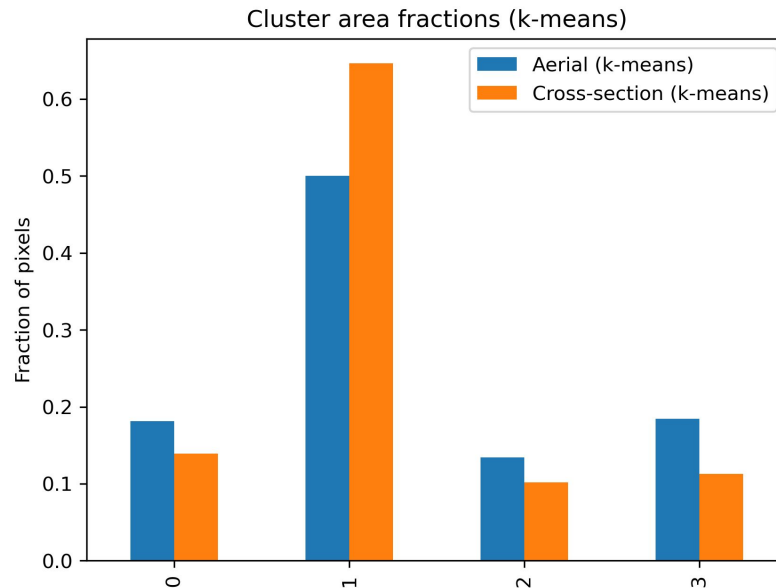
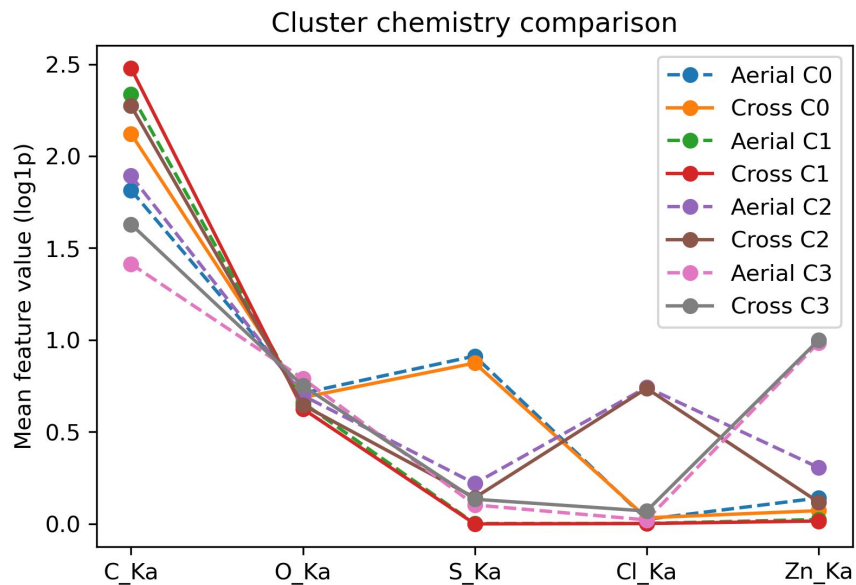
$$RI(P, G) = \frac{a + d}{a + b + c + d}$$

$$ARI = \frac{RI - E(RI)}{1 - E(RI)}$$

- High agreement between methods suggesting robust, meaningful segmentation



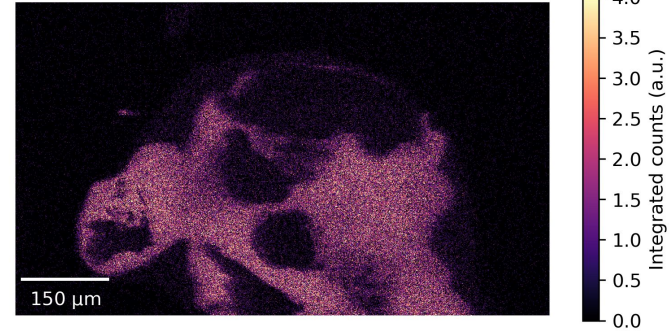
Aerial vs cross-section comparison



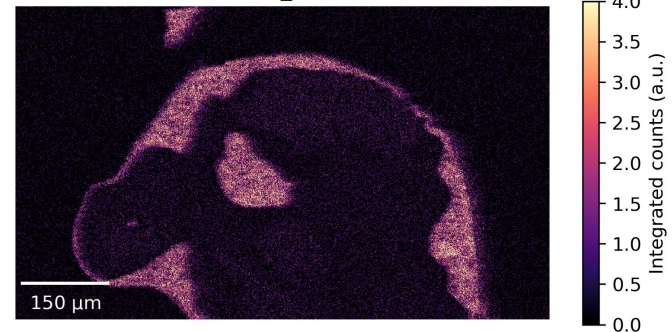
Conclusions

- Clustering identified chemically meaningful regions
- Four clusters consistently identified
- Strong agreement between algorithms and geometries

Aerial: Zn_Ka map



Aerial: S_Ka map



Next steps & future directions

Next steps →

- continue optimizing preprocessing
- validating clusters using independent point-based EDS spectra
- include additional elemental features (Cd)

Future directions →

- testing additional dimensionality-reduction methods
- incorporating additional spectral features
- applying the workflow to other hyperspectral imaging datasets

Thank you!

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Supplemental slides:

Machine Learning in Heritage Science

- Applied to a broad range of topics
 - Pigment analysis, damage detection, virtual restoration, damage prediction, etc.
- Rich results on **hyperspectral datasets** (VIS, UV, IRR) and XRF

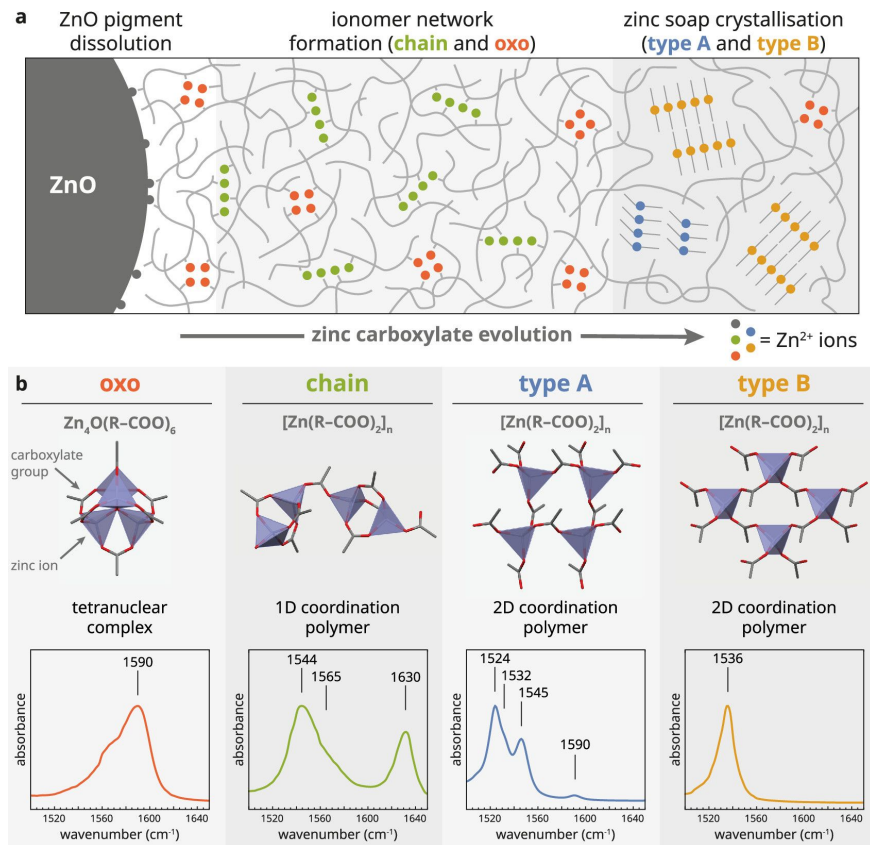
Table 1 | An overview of state-of-the-art studies published in English-language scientific journals that apply ML to painting conservation

	Authors	Data source	Input data	Type of learning	Algorithm	Output
Enhancing the Readability of Scientific Imagery	Karella, Blažek, and Striova ¹²	Artificial data (virtual phantoms), modern data (mock-ups), Historical data (paintings)	VIS-NIR RIS data	Supervised learning	Modern deep learning	Enhanced infrared reflectogram
	Pu et al. ¹⁶	Artificial data (mixed X-ray and RGB image), Historical data (painting)	VIS, XRR data	Self-supervised learning	Classic neural network, Modern deep learning	Separated X-ray
Pigment Analysis	Jones et al. ²¹	Artificial data (XRF spectra with FP), historical data (painting)	XRF data	Supervised learning	Modern deep learning	Pigment identification
	Xu et al. ²⁴	Artificial data (XRF spectra with FP), modern data (mock-ups), historical data (paintings)	XRF data, RIS data	Supervised learning	Modern deep learning	Pigment identification
	Radpour et al. ²⁰	Historical data (paintings)	RIS data	Supervised learning	Modern deep learning	Pigment identification
	Pouyet et al. ²³	Modern data (pure pigment pellets, mock-ups), historical data (paintings)	SWIR hyperspectral datasets	Supervised learning	Classic neural network	Pigment unmixing
	Vermeulen et al. ⁴⁴	Modern data (mock-up), Historical data (painting)	XRF data, RIS data	Unsupervised learning	Traditional ML algorithm (K-SVD)	Pigment unmixing
	Shitomi et al. ²⁵	Artificial data (spectral data with KM), modern data (mock-ups)	Artificial spectral data, VIS data	Unsupervised learning	Classic neural network	Pigment unmixing, estimated pigment layer thickness
Damage Detection	Sidorov and Hardeberg ⁴⁵	Historical data (paintings)	Greyscale images	Supervised learning	Modern deep learning	Crack map
	Sindel, Maier, and Christlein ⁴⁶	Historical data (paintings)	VIS, UV, IRR, XRR images	Supervised learning	Modern deep learning	Crack map
	Nadisic et al. ²⁹	Historical data (paintings)	VIS, IRP, IRR, XRR images	Supervised learning	Modern deep learning	Crack map
	Meeus et al. ²³	Historical data (paintings)	VIS, IRP, IRR, XRR images	Semi-supervised learning	Modern deep learning	Paint loss map
	Chirosca and Angheluta ¹	Historical data (paintings)	VIS images	Supervised learning	Modern deep learning	Labelled damage map
	Liu et al. ³³	Artificial data (simulated damages), Historical data (paintings)	IRT images	Unsupervised learning	Generative Adversarial Network, Modern deep learning	Damage map
	Moradi et al. ⁴⁷	Modern data (mock-up)	IRT images	Supervised learning	Classic neural network, Modern deep learning	Damage map
Virtual Restoration	Sizyakin et al. ³⁵	Historical data (paintings)	VIS, IRP, IRR, XRR images	Supervised learning	Generative Adversarial Network	Virtually restored image
	Sun et al. ³⁶	Artificial data (simulated damages), historical data (paintings)	VIS images	Supervised	Generative Adversarial Network	Virtually restored image
	Maali Amiri and Messinger ²⁷	Artificial data (artificially yellowed hyperspectral image), historical data (paintings)	Hyperspectral images	Unsupervised learning	Modern deep learning	Virtually restored image
	Bombini et al. ³⁹	Artificial data (XRF spectra with ganX)	XRF data	Supervised learning	Modern deep learning	Virtually restored image
Damage Prediction	Califano et al. ⁴²	Artificial data (2D submodel of FE simulations)	Digital 3D model	Supervised learning	Traditional ML algorithm	Classification graph

Vijle, Aster Van, Hacigüzeller, Piraye, and Snickt, Geert Van der, Machine learning for painting conservation: a state-of-the-art review, npj Heritage Science, n.d., 1, 13. <https://doi.org/10.1038/s40494-025-01924-3> (accessed 2/20/2026).

Zinc carboxylate (zinc soap) formation

- Hydrolysis of the oil binder releases free fatty acids from triglycerides during aging
- Zn^{2+} ions from pigments (ex: Zinc Oxide) diffuse into polymerized oil matrix
- Fatty acids coordinate with Zn^{2+} , forming insoluble zinc carboxylates (zinc soaps) that aggregate within the paint layer

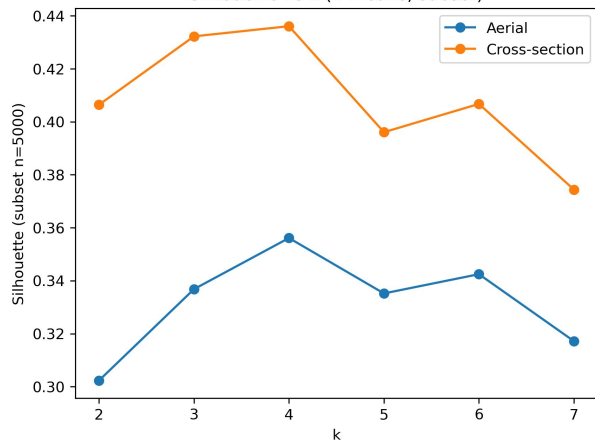


ACS Appl. Polym. Mater. 2020, 2, 12, 5674-5685

Cluster k-value metrics

Silhouette

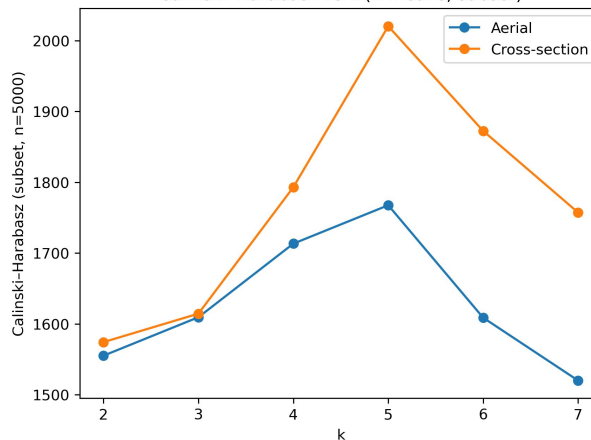
Silhouette vs k (k-means, subset)



$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

Calinski-Harabasz

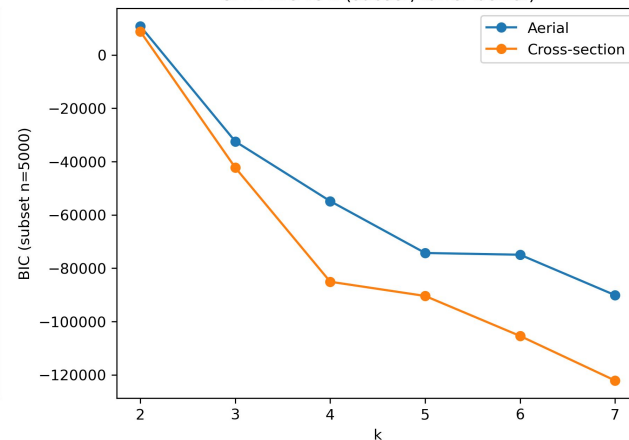
Calinski-Harabasz vs k (k-means, subset)



$$CH = \frac{\text{between-cluster variance}}{\text{within-cluster variance}} \times \frac{n - k}{k - 1}$$

BIC

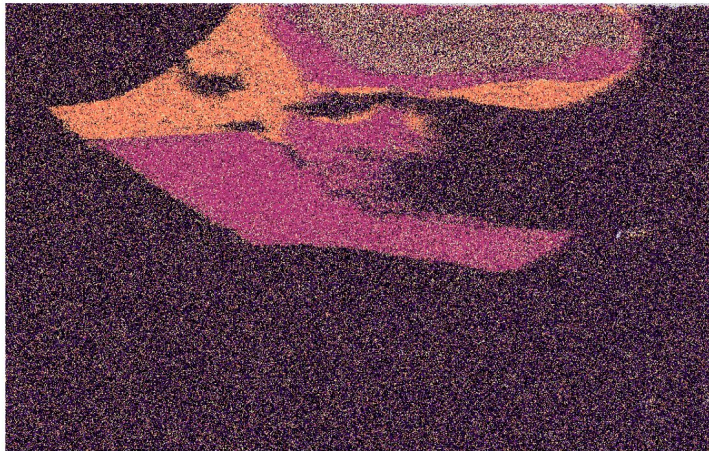
GMM BIC vs k (subset; lower better)



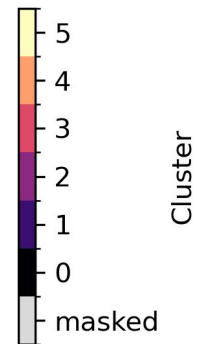
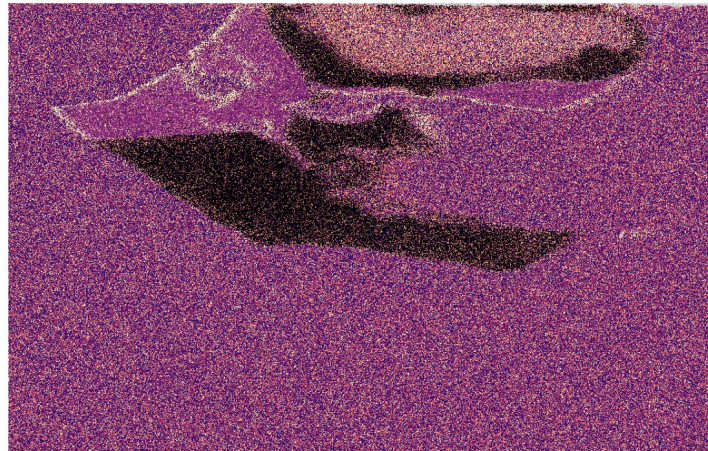
$$BIC = -2 \ln(L) + p \ln(n)$$

Cluster k-value sweep

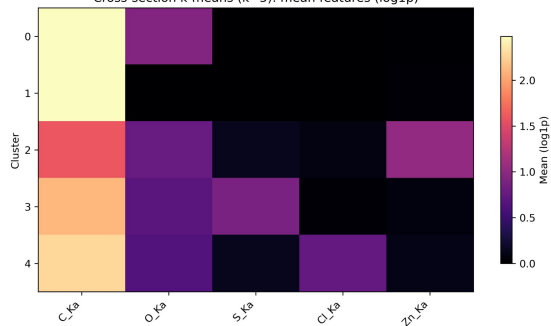
Cross-section k-means (k=5)



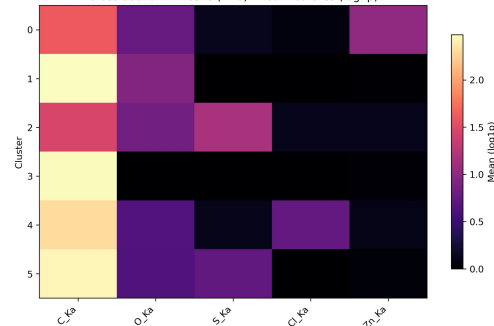
Cross-section k-means (k=6)



Cross-section k-means (k=5): mean features (log1p)

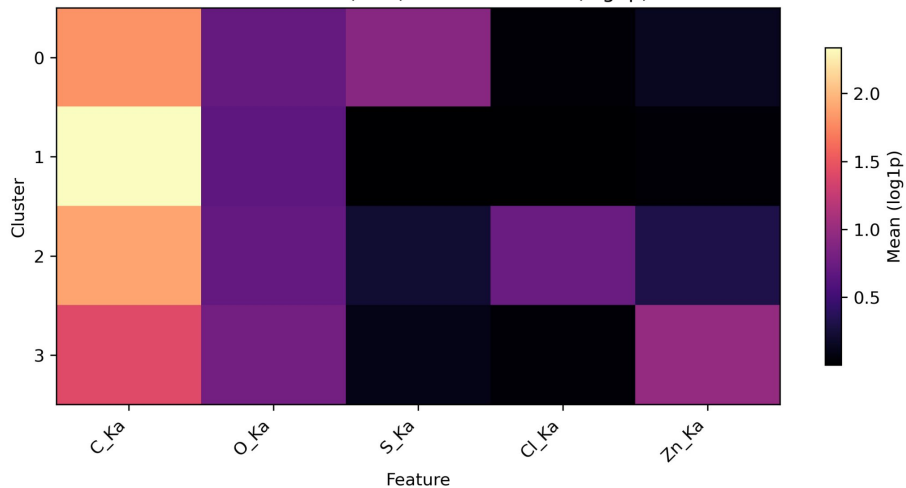


Cross-section k-means (k=6): mean features (log1p)



K-Means Output (cluster “breakdown”)

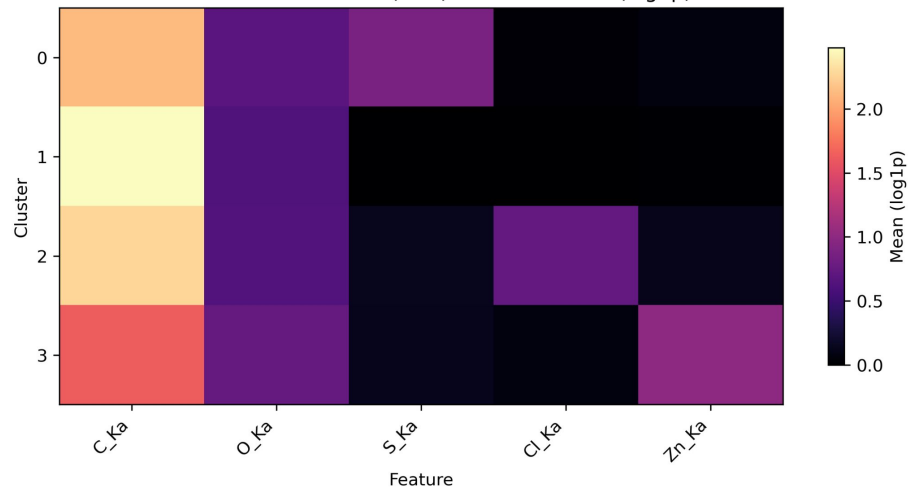
Aerial k-means (k=4): mean features (log1p)



Aerial k-means (k=4) cluster mean features (log1p):

cluster	C_Ka	O_Ka	S_Ka	Cl_Ka	Zn_Ka
0	1.815	0.711	0.913	0.025	0.140
1	2.338	0.659	0.002	0.002	0.024
2	1.893	0.698	0.220	0.743	0.307
3	1.413	0.790	0.103	0.022	0.985

Cross-section k-means (k=4): mean features (log1p)

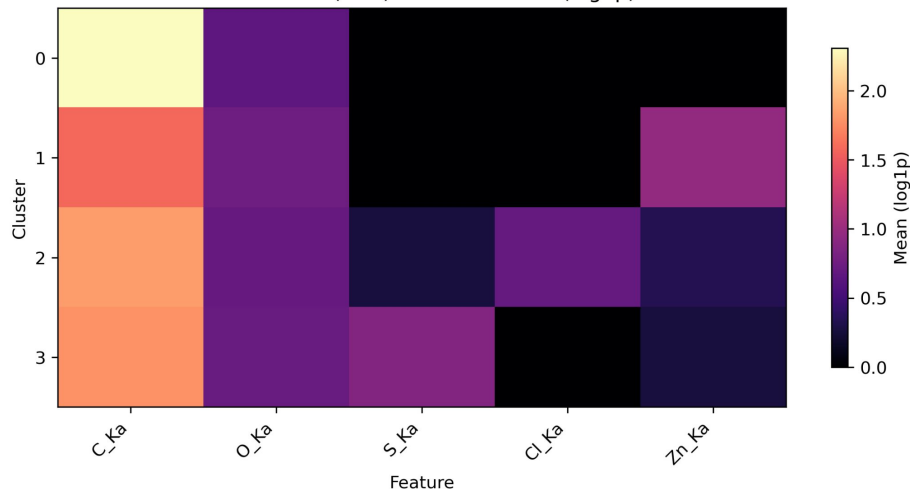


Cross-section k-means (k=4) cluster mean features (log1p):

cluster	C_Ka	O_Ka	S_Ka	Cl_Ka	Zn_Ka
0	2.121	0.687	0.875	0.033	0.072
1	2.479	0.626	0.000	0.002	0.016
2	2.273	0.647	0.142	0.737	0.119
3	1.629	0.751	0.134	0.070	0.998

GMM Output (chemical “breakdown”)

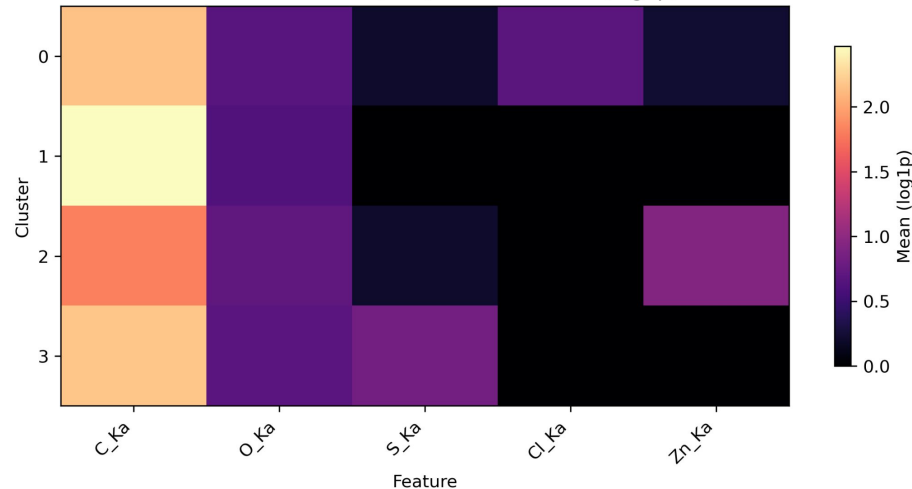
Aerial GMM (k=4): mean features (log1p)



Aerial GMM (k=4) cluster mean features (log1p):

cluster	C_Ka	O_Ka	S_Ka	Cl_Ka	Zn_Ka
0	2.308	0.666	0.000	0.000	0.000
1	1.587	0.752	0.000	0.000	0.959
2	1.837	0.709	0.268	0.694	0.341
3	1.776	0.722	0.869	0.000	0.267

Cross-section GMM (k=4): mean features (log1p)

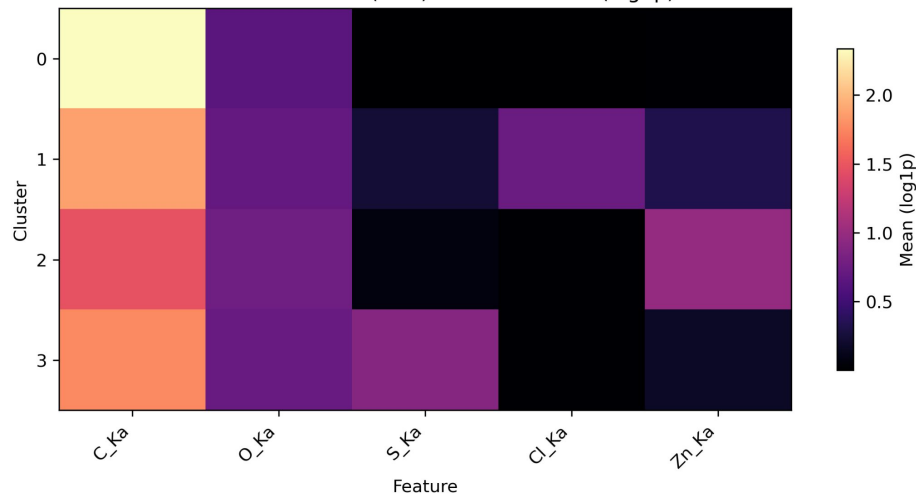


Cross-section GMM (k=4) cluster mean features (log1p):

cluster	C_Ka	O_Ka	S_Ka	Cl_Ka	Zn_Ka
0	2.155	0.671	0.211	0.693	0.226
1	2.468	0.627	0.000	0.000	0.000
2	1.821	0.719	0.203	0.000	0.928
3	2.170	0.680	0.838	0.000	0.000

Hierarchical Output (chemical “breakdown”)

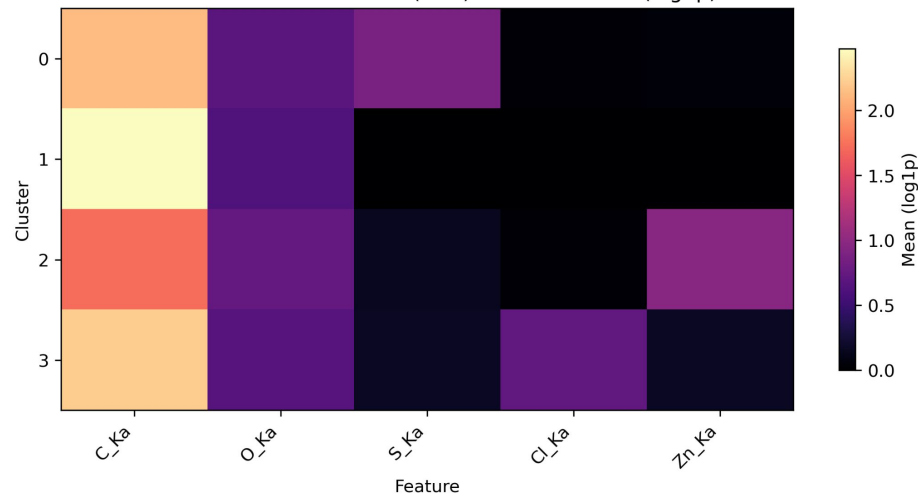
Aerial hierarchical (k=4): mean features (log1p)



Aerial hierarchical (k=4) cluster mean features (log1p):

cluster	C_Ka	O_Ka	S_Ka	Cl_Ka	Zn_Ka
0	2.337	0.657	0.007	0.002	0.012
1	1.873	0.702	0.238	0.741	0.321
2	1.479	0.776	0.071	0.013	0.981
3	1.769	0.729	0.905	0.016	0.172

Cross-section hierarchical (k=4): mean features (log1p)



Cross-section hierarchical (k=4) cluster mean features (log1p):

cluster	C_Ka	O_Ka	S_Ka	Cl_Ka	Zn_Ka
0	2.477	0.627	0.000	0.002	0.006
1	2.222	0.658	0.169	0.735	0.160
2	1.723	0.728	0.146	0.038	0.961
3	2.135	0.687	0.876	0.021	0.046